

Amazon Internship Interview Experience

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About the company

Amazon is a global tech company that started as an online bookstore and grew into one of the largest e-commerce platforms. It lead in cloud computing with AWS, innovate with devices like Echo, and focus on customer obsession, operational excellence, and continuous innovation.

Role details Offered role:

SDE Intern (Software Development Engineer)

Tenure: 2 months

Preliminary Round

The preliminary round was conducted online using Amazon's own coding platform. In this round, there were 2 coding questions that needed to be solved within 1.5 hours. While I don't recall the exact wording of the questions, the first question was like

You are given a list of product names as strings, representing the order in which customers have interacted with them on Amazon's platform. You need to determine the order of these product names based on their frequency of occurrence. If two product names have the same frequency, the one that appears second or more times before the other should come first. Write a function that returns a list of product names sorted by these criteria.

Input:

A list of strings where each string represents a product name.

Output:

A list of strings sorted by frequency of occurrence. If two product names have the same frequency, the one that appears for the last time before the other in the input list should appear first in the output list.

Example:

Input: ["Laptop", "Phone", "Tablet", "Laptop", "Phone", "Tablet", "Tablet"]

Output: ["Tablet", "Laptop", "Phone"]

Explanation:

- "Tablet" appears 3 times.
- "Laptop" appears 2 times.

- "Phone" appears 2 times.

Since "Tablet" has the highest frequency, it appears first. "Laptop" and "Phone" have the same frequency, but "Tablet" appears before "Phone" for the second time, so "Tablet" comes before "Phone" in the output.

The second question was related to some mathematical operation with the string. I can't remember the exact question and it was a medium level question.

I managed to solve both coding problems in the preliminary round within 20-25 minutes. After the coding test, there was a mandatory work-level assessment, consisting of around 60 questions focused on Amazon's Leadership Principles. Many candidates mentioned that they completed both coding problems but chose to skip this work-level assessment, which resulted in them not being shortlisted. Therefore, it is crucial to complete the work-level assessment to proceed in the selection process.

For more information on Amazon's Leadership Principles, you can refer to this [link](#).

Technical Interview:

Around 400 students attended the online test and out of that only 20 students got selected for the next round.

We received a 1-hour time slot for an online interview the day before the interview, which took place on Amazon Chime. At the start of the interview, the interviewer sent a code editor link via email, where he pasted the question. I was required to write my solution directly in that code editor.

The interviewer, an SDE-II from Amazon's Bangalore office, briefly introduced himself and mentioned that he had reviewed my resume, so he did not need an introduction from me. The interview began with a question related to a payment system project I had worked on. After explaining my solution, he seemed satisfied. He then casually asked how my day was going, to which I mentioned that I had a compiler exam that morning. This led to questions about the usefulness of studying compilers, which I answered by explaining their relevance.

The interviewer then moved on to the coding question, which was a variation of the three-sum problem. The task was to return all unique combinations, even if there were repetitive elements. I first explained the brute-force approach and then moved on to the optimal solution, detailing the time and space complexities for each. After the explanation, he asked me to write the code in the code editor, and we conducted a dry run together. He also inquired about the time complexities of operations like insertion, search, and delete in map and set.

During the interview, the interviewer briefly asked me to explain Binary Search Trees (BST). He then gave me a BST-related question: I was asked to perform preprocessing on a given BST so that after this preprocessing, a preorder traversal would return the values in descending order. I explained and implemented a solution using the mirror algorithm. [Reference](#). The interviewer was surprised that I solved it within 5 minutes. He then asked about the time complexity of this recursive solution, followed by questions about space complexity, such as the number of stacks created, the worst-case conditions, and the number

of stacks if the tree is skewed. We discussed further details related to both time and space complexities.

At the end of the interview, he asked if I had any questions. I inquired about the work culture at Amazon and asked him about his personal experience working there.

Result: The following afternoon, I was preparing for my Lab exam when, around 11 am, I received news from CUIC that out of 20 candidates, only 2 had been shortlisted, and I was one of them. I was so happy about the result that I almost forgot I had an exam that afternoon! 😊

Tips:

- Always start by explaining the brute-force solution, even if you already know the optimal approach as they will assess your thought process too.
- Stay engaged during the interview; avoid being silent.

Preparation Docs:

For my preparation, I personally followed Striver's A2Z DSA sheet, which you can find [here](#). This resource helped me understand and cover most of the patterns in DSA. After learning each new topic, I also solved random questions from LeetCode and GeeksforGeeks to reinforce my understanding. Additionally, I followed Abdul Bari's DSA Algorithms playlist, which provided a solid foundation in various algorithms.

Instead of focusing on solving a large number of similar problems, I recommend trying to solve a variety of different patterns. This approach not only prepares you for unexpected questions but also builds a deeper understanding of DSA concepts.

All the very best! Feel free to reach out to me if you have any doubts.

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